

Original Article

Integrating host use and dispersal ability with species delimitation to unravel a cryptic radiation of photosynthetic sea slugs

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ABSTRACT

Species delimitation models are important quantitative tools for facilitating the discovery of cryptic species and clarifying the boundaries between focal taxa. However, distinct evolutionary processes may promote diversification at different points in a radiation, complicating efforts to identify speciation processes and diagnostic characters. Moreover, traits such as dispersal ability can have opposing effects on diversification rate versus population subdivision, which causes coalescent models to oversplit samples. We examined the effects of larval dispersal and host use on population structure and diversity in a Caribbean radiation of photosynthetic herbivores, sea slugs in the genus *Elysia*. Five methods of species delimitation broadly supported six described taxa and a previously unrecognized taxon, but species with less dispersive offspring were more likely to be oversplit using the multispecies coalescent. Surprisingly, species that shared a phylogeographic break were likely to be split into multiple taxa by coalescent models regardless of dispersal ability. Some host shifts may have contributed to divergence, but others pre-dated speciation, suggesting ecological specialization can promote reproductive isolation but often acted at deeper timescales; recent speciation was more often driven by sea level fluctuations and changes in reproductive anatomy, potentially reinforced by dispersal limitation.

Keywords: Caribbean; DNA barcoding; ecological speciation; *Elysia*; host shifts; multispecies coalescent; phylogeography; Sacoglossa; sexual selection

INTRODUCTION

From the earliest phase of molecular taxonomy and DNA barcoding, species delimitation models (SDMs) have provided insights into evolutionary processes that generate previously unrecognized biodiversity (Hubert and Hanner 2015, Miralles *et al.* 2024). The inherently hidden nature of cryptic species can even reveal processes that may be otherwise overlooked (Marki *et al.* 2018, Struck *et al.* 2018, De Jode *et al.* 2019, Cheng *et al.* 2025). However, SDMs may not easily differentiate candidate species and potential mechanisms of speciation when multiple evolutionary processes drive or reinforce divergence among populations as well as species (Dellicour and Flot 2015, Luo *et al.* 2018, Leaché *et al.* 2019, Sukumaran *et al.* 2021). Lineages often covary in multiple traits that could contribute to divergence scenarios, such as dispersal and

range size; host or habitat use; and/or reproductive anatomy or behaviour (e.g. sexually selected traits). Complex evolutionary scenarios can result when such traits act in concert, or at different times in the history of a radiation; it may become computationally prohibitive to compare all the alternative models to identify the process that best correlates with speciation, and not older splits or shallower population subdivision (Smith and Carstens 2020).

A further complication is that trait-dependent diversification can result from character change that affects dispersal (Burridge and Waters 2020), niche shifts (Blaimer *et al.* 2023, Cornwallis *et al.* 2023), outcrossing (Goldberg *et al.* 2010), or sexual signalling (Brock *et al.* 2021). In principle, species delimitation should be a first step for testing theory regarding so-called species selection, with SDMs useful in identifying bursts of diversification.

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However, the same traits often affect emergent properties of species (e.g. rates of gene flow; range size) that lower the accuracy of SDMs (Wessinger 2021, Suárez *et al.* 2022). Notably, multispecies coalescent models oversplit when populations are genetically subdivided (Sukumaran and Knowles 2017, Leaché *et al.* 2019) or widely distributed and experiencing isolation-by-distance (Chambers and Hillis 2020, Mason *et al.* 2020). A trait that increases genetic structure will probably cause such SDMs to oversplit the resulting lineage (Huang 2020, Mirarab *et al.* 2021), which complicates efforts to test the effects of that trait on diversification. Thus, we cannot analyse trait effects on evolutionary processes without delimiting species, but the same trait can affect diversification and also bias SDMs, potentially in opposing ways.

DNA barcoding surveys of marine invertebrate groups usually reveal cryptic taxa, improving biodiversity estimates and informing conservation efforts for threatened reef hotspots (Cerca *et al.* 2020, Elias-Gutierrez *et al.* 2021, Mugnai *et al.* 2021, Maslakova *et al.* 2022). However, such taxa pose interesting challenges for SDMs due to the layering effects of rapidly evolving traits that contribute to dispersal, habitat choice, and mate preference, and which may therefore promote diversification at a range of evolutionary depths. For instance, benthic invertebrates produce either planktotrophic (feeding, highly dispersive) or lecithotrophic (non-feeding, less dispersive) larvae, with cascading effects on population persistence, genetic structure and speciation–extinction dynamics (Collin and Moran 2018, Wray 2020). Development mode became a textbook example of species selection after palaeontological studies suggested lecithotrophy promoted speciation, although later analyses indicated planktotrophy increases diversification by buffering against extinction (Krug *et al.* 2015, Ellingson and Krug 2016, Moreno *et al.* 2023). Testing these alternative hypotheses is challenging, however, because lecithotrophic species also have less gene flow and more structured populations (Weersing and Toonen 2009, Kelly and Palumbi 2010, Selkoe and Toonen 2011, Dawson *et al.* 2014, Marko and Hart 2018). Lecithotrophic taxa are likely to be oversplit by coalescent SDMs, appearing as clusters of young species rather than as subdivided populations; subsequent comparative analyses would then be biased towards detecting increased diversification driven by lecithotrophy, even if this larval type actually lowered diversification rate at macroevolutionary timescales.

In addition, some clades of marine animals have strong host, habitat, or depth preference traits. Such associations can lead to the formation of ecotypes, which through disruptive selection may evolve into specialists partitioned across an environmental gradient or among available hosts or habitat features (González *et al.* 2018, Titus *et al.* 2019). Such traits may reduce distributions or population sizes, which could bias SDM analyses, but also introduce opportunities for ecological speciation (Bowen *et al.* 2016, Potkamp and Fransen 2019, Holt *et al.* 2020, Prada and Hellberg 2021, Hernández-Hernández *et al.* 2021). Dispersal ability and host specificity may also be correlated (Matthews *et al.* 2023); trait covariance that reinforces to subdivide populations may be a particular source of bias for SDMs. Reproductive compatibility and sexual selection can also contribute to speciation, including during secondary contact (Kopp *et al.* 2018, Foote 2018). For instance, marine animals have rapidly evolving gamete recognition proteins (Carlisle and Swanson 2021) and anatomical structures (Moreno

et al. 2023) that experience strong selection and rapidly change effective population sizes.

Here, we examine how changes in dispersal, host, or sexually selected traits affect phylogeographic structure and species diversity assessed by SDMs for a Caribbean radiation of herbivorous sea slugs. *Elysia* are photosynthetic heterobranchs (order Sacoglossa) that retain functional chloroplasts from their algal diet for weeks, or months in a few cases (Händeler *et al.* 2009, Cruz and Cartaxana 2022, Morelli *et al.* 2024). Most elysiids specialize on one algal genus or a few preferred species, and host shifting has been proposed as a mechanism for ecological speciation in *Elysia* (Rodríguez and Krug 2022) and other sacoglossans (Berriman *et al.* 2018, Krug *et al.* 2018a, 2018b, Moreno *et al.* 2023). Sister species of *Elysia* also frequently differ in larval type, which can also drive the rapid evolution of reproductive isolation; planktotrophy was correlated with higher diversification rates in this group, contrary to long-standing theory based on the fossil record of gastropods (Krug 2009, Krug *et al.* 2015, Ellingson and Krug 2016). Some elysiids use a sharp penial stylet for hypodermic insemination, but most reciprocally inseminate with an unarmed penis (Krug *et al.* 2016). Thus, several evolutionary forces may promote diversification in this group, including changes in host, larval type, and penial armature (Krug 2011). However, the same traits and forces can impact population genetic structure and hence SDM performance, warranting careful attention for systematic revisions.

The Western Atlantic holds about a quarter of elysiid diversity, including several endemic radiations (Lugo *et al.* 2025). However, systematists disagree about the extent of that diversity and the identities of species (Krug *et al.* 2016, Ortea Rato *et al.* 2005, Ortea 2018). To resolve the number of species, and identify evolutionary processes promoting marine radiations, we analysed intra-specific phylogeographic structure and delimited species in the *E. papillosa* complex (Fig. 1). We modelled the evolutionary history of algal host associations, and compared taxa with dispersive versus non-dispersive larvae, and with or without a penial stylet. Our findings show alternative evolutionary processes promoted divergence at different stages in the history of this radiation, mediated by traits in some cases and probably by environmental change in others. We discuss the performance of SDMs in light of the complex history of forces acting on this lineage, and their power to reveal previously undetected candidate species and provide insight into the speciation process.

MATERIALS AND METHODS

Sample collection and host identification

We analysed all available specimens and sequence data for specimens belonging to the *E. papillosa* complex, termed subclade 2 in the most recent analysis of the family Plakobranchidae (Lugo *et al.* 2025). Six species were recognized in this clade by Krug *et al.* (2016), including three taxa that were described based on genetic distance from congeners and internal morphological differences (Fig. 1). However, Krug *et al.* (2016) did not analyse range-wide genetic data or apply SDMs. The validity of *E. taino* was subsequently questioned by Ortea (2018), who resurrected *E. annedupontae* and named a highly similar species, *E. frankenstein*, with no supporting morphological or ecological data.

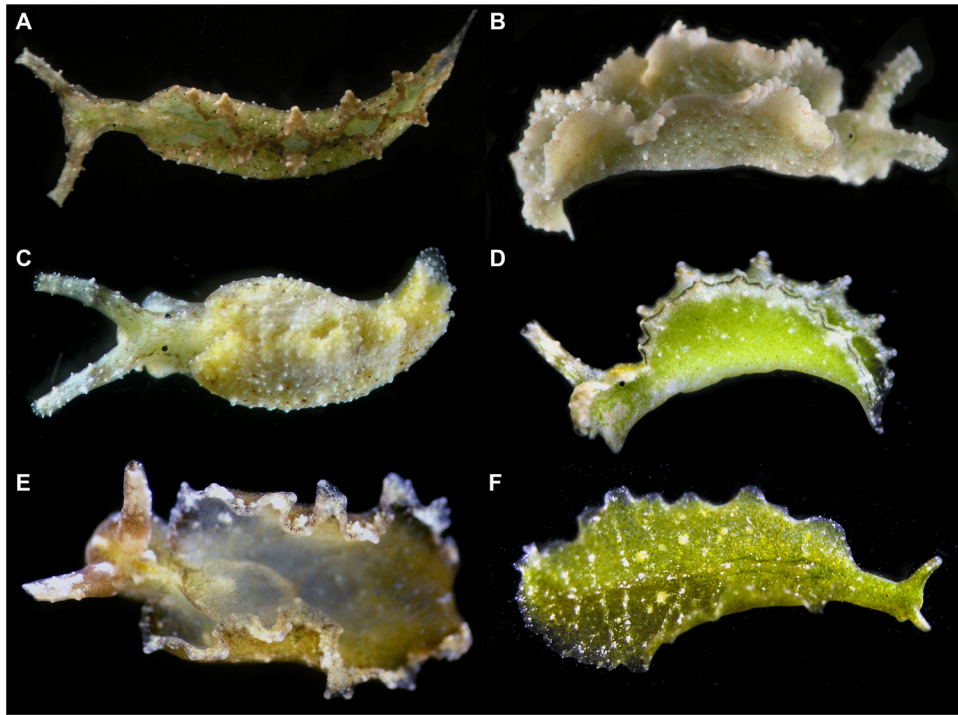


Figure 1. Live specimens of *Elysia* spp. from subclade 2. A, *E. papillosa* from Bocas del Toro, Panama; actual size of specimen = 7 mm. B, *E. taino* from Discovery Bay, Jamaica; actual size of specimen = 5 mm. C, *E. patina* from Sweetings Cay, Grand Bahamas Island, Bahamas; actual size of specimen = 6 mm. D, *E. zuleicae* from Discovery Bay, Jamaica (isolate Ezul_06JAM05); actual size of specimen = 5 mm. E, *E. buonoi* from San Salvador, Bahamas (isolate Ebuo_07SSAL10); actual size of specimen = 6 mm. F, *E. christinae* from Bimini, Bahamas (isolate Echr_10BIM02); actual size of specimen = 9 mm.

Samples were collected by the authors from 18 western Atlantic sites over 12 years with permission of the State of Florida or host country, except samples from Dominica and Antigua (collected by D. Willette) and the U.S. Virgin Islands (USVI; collected by Á. Valdes) (Fig 2; Table 1). To assess host algal use, each udotacean green alga (0.5–1 kg wet weight) was separately collected at every site by removing algal stipes from the substrate, placing algae in a sealed bag, and transferring to an aerated bin of seawater. Slugs were removed and provisionally identified under a stereomicroscope, then photographed and preserved in 100% ethanol. Host use was inferred from field distributions and confirmed by feeding observations in the laboratory.

Phylogenetic analyses of interspecific relationships

Genomic DNA was extracted with DNeasy® kits (Qiagen) from tissue dissected from the posterior end of 353 specimens (Table 2). Portions of two loci were amplified by polymerase chain reaction (PCR) using universal primers: the mitochondrial cytochrome *c* oxidase I (*COI*) gene and the nuclear histone III (*H3*) gene; amplification and sequencing used described reaction conditions (Krug *et al.* 2015, 2018b). Chromatograms were edited in GeneiousPro 4.8 software and aligned with MAFFT, yielding up to 658 bp for *COI* and 328 bp for *H3* (Supporting Information Table S1). Identical *COI* haplotypes were collapsed, yielding 234 unique ingroup sequences that were deposited in the NCBI database (Table S1). For a given analysis, shorter *COI* haplotypes were either extended with question marks to denote missing data (phylogenetic analyses) or all were trimmed to the length of the shortest sequence

obtained [analysis of molecular variance (AMOVA), haplotype networks]. Chromatograms for *H3* were examined for heterozygous positions, and allelic phase was resolved by subtracting the common allele from heterozygotes to identify rare alleles, or where necessary by next-generation sequencing by synthesis.

Evolutionary relationships for *COI* haplotypes were inferred using maximum likelihood (ML), and Bayesian inference (BI) with Markov-chain Monte Carlo (MCMC). Outgroups were designated based on a four-gene analysis of 154 species in the family Plakobranchidae (Lugo *et al.* 2025), including exemplars of *E. pusilla*, *E. stylifera*, and the undescribed *Elysia* sp. 26. ML analysis was implemented in RAxML v.7.2.7 (Stamatakis 2006) through the CIPRES web portal (Miller *et al.* 2010), parameterizing a GTR+ Γ model, using a separate partition for the fast-evolving third codon position. Bootstrap support (BS) $\geq 70\%$ was deemed significant (Douady *et al.* 2003), with the number of pseudoreplicates determined during the run.

For BI analyses, mixture models of sequence evolution were implemented in *BayesPhylogenies*. The best-fit model was assigned to each position using a likelihood criterion without a priori partitioning (Pagel and Meade 2004). Four replicate Markov chains were run for 2×10^7 generations each, sampling every 5×10^3 generations. A reverse-jump implementation of the MCMC sampler was first used to estimate the number of models needed to capture among-site heterogeneity in evolutionary rates and base composition; based on those results, each chain was run with three GTR+ Γ (four rate classes) models. Posterior distributions were visualized in Tracer v.1.7 (Rambaut *et al.* 2018) to confirm stationarity and

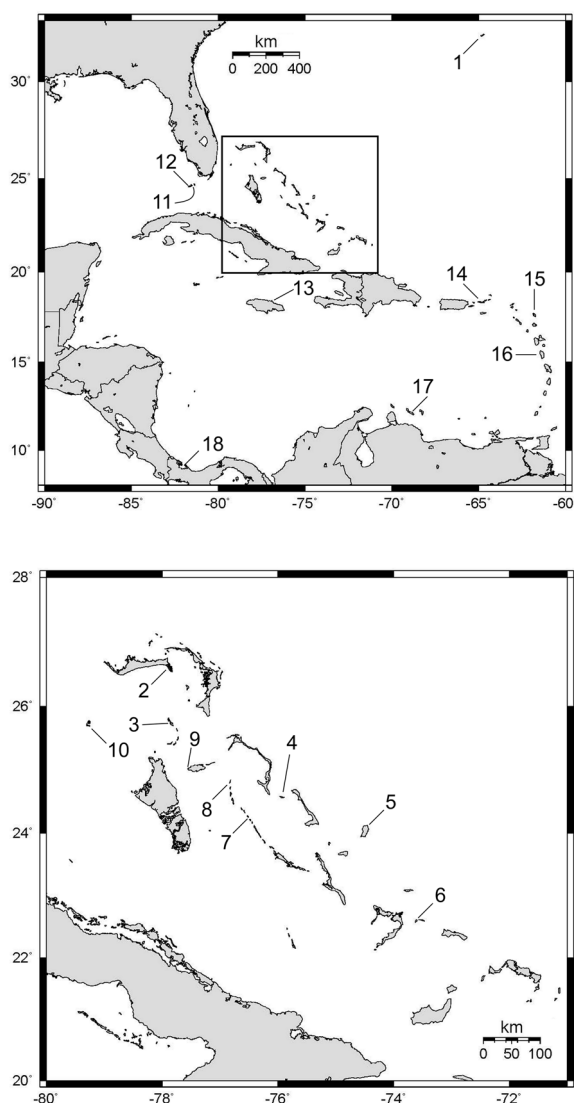


Figure 2. Sampling locations for sequenced specimens; coordinates and sampling dates are given in Table 1.

convergence; the final 25% of all runs (10^3 trees per chain) was pooled as the post-burnin sample. A 50% majority-rule consensus tree was calculated in *BayesTrees* (<http://www.evolution.reading.ac.uk>) with mean branch lengths and posterior probability (Pp) of nodes, taking $Pp \geq 90\%$ as significant (Huelsenbeck and Rannala 2004, Simmons et al. 2004).

Species delimitation

Four methods of molecular species delimitation were performed on collapsed COI haplotypes excluding outgroups. First, a matrix of TrN-corrected pairwise distances was calculated in Mega v.1.1 (Tamura et al. 2021) and uploaded to the webtool (<https://bioinfo.mnhn.fr/abi/public/abgd/abgdold.html>) for Automated Barcode Gap Discovery (ABGD; Puillandre et al. 2012). Priors were set to $pmin = 0.005$, $pmax = 0.08$, and modified minimum relative gap width < 1.3 , based on prior results. The number of supported species and associated barcode gap were analysed across the range of plausible values for the prior intraspecific divergence. Second, we ran Assemble Species by Automatic

Table 1. Collection localities, codes, and dates for sampled *Elysia* spp. from the Western Atlantic.

Site	Latitude, longitude	Collection dates
1. Bermuda, Whale Bone Bay (Ber)	32°21'53"N, 64°42'46"W	6/2006
Bahamas		
2. Sweetings Cay (Swe)	26°33'43"N, 77°51'15"W	6/2004, 7/2007, 7/2010
3. Great Stirrup Cay (Stir)	25°49'12"N, 77°53'56"W	7/2007
4. Little San Salvador (LSS)	24°34'30"N, 75°56'30"W	6/2004, 7/2007
5. San Salvador (SSal)	24°08'30"N, 74°28'18"W	6/2004, 7/2007, 7/2010
6. Plana Cays (Pla)	22°36'40"N, 73°33'52"W	6/2004, 7/2007
7. Compass Cay (Comp)	24°16'29"N, 76°30'35"W	7/2010
8. Northern Exumas (NEx)	24°46'00"N, 76°49'01"W	7/2010
9. New Providence (NPr)	25°00'35"N, 77°34'00"W	7/2010
10. Bimini (Bim)	25°44'19"N, 79°16'20"W	7/2010
Florida Keys, USA		
11. Mote Marine Lab (Mot)	24°39'42"N, 81°27'16"W	6/2007, 5/2013
12. Geiger Beach, Key West (Gei)	24°33'39"N, 81°40'39"W	7/2006, 6/2007, 10/2009
13. Jamaica, Discovery Bay (Jam)	18°28'07"N, 77°24'53"W	3/2006
14. U.S. Virgin Islands, St. Thomas (USVI)	18°21'N, 64°52'W	2014
15. Antigua, Parham Bay (Ant)	17°06'45"N, 61°45'43"W	4/2008
16. Dominica, Anse de Mai (Dom)	15°34'06"N, 61°27'56"W	6/2007
17. Curaçao (Cur)		1/2009
Spanish Harbor	12°05'07"N, 68°50'55"W	
Piscadero Bay	12°08'07"N, 68°57'54"W	
18. Panama, Bocas del Toro (Pan)	09°19'60"N, 82°15'00"W	12/2004, 7/2015

Partitioning (ASAP; Puillandre et al. 2021), using the webtool (<https://bioinfo.mnhn.fr/abi/public/asap/asapweb.html>) to calculate a matrix of K2P distances from the same COI alignment, reporting the number of species supported by the highest ASAP score and the associated barcode gap.

Next, we used the General Mixed Yule-Coalescent model (GMYC) to identify the threshold on a tree where the splitting process is better modelled as a speciation versus a coalescent process (Pons et al. 2006, Fujisawa and Barraclough 2013). We generated ultrametric COI haplotype trees in BEAST2 (Bouckaert et al. 2019) with substitutions modelled as a GTR+ Γ process (four rate categories), under four combinations of a relaxed clock (log normal) or a strict (constant) clock, and a Yule or a

Table 2. Collection localities and number of specimens sampled for Caribbean *Elysia* spp.

Site	<i>E. papillosa</i>	<i>E. taino</i>	<i>E. zuleicae</i>	<i>E. cf. zuleicae</i>	<i>E. buanoi</i>	<i>E. patina</i>	<i>E. christinae</i>
1. Bermuda (Ber)	16		8				
Bahamas							
2. Sweetings Cay (Swe)	16	1	11			11	
3. Great Stirrup Cay (Stir)	3	7	12				
4. Little San Salvador (LSS)	17	3	17				
5. San Salvador (SSal)	1	6	15		4		
6. Plana Cays (Pla)	2	8	10				
7. Compass Cay (Comp)	1	3					
8. Northern Exumas (NEx)	1	2	14				
9. New Providence (NPr)		5					
10. Bimini (Bim)	10	1	13			6	2 (+8) ^a
Florida Keys, USA (FL)							
11. Mote Marine Lab (Mot)						4	
12. Geiger Beach, Key West (Gei)	23		9				
13. Jamaica, Discovery Bay (Jam)	2	13	11				
14. U.S. Virgin Islands, St. Thomas (USVI)	1	1	13	1			
15. Antigua (Ant)	1						
16. Dominica (Dom)		7					
17. Curaçao (Cur)	8		11				
18. Panama, Bocas del Toro (Pan)	12		11				
N =	114	57	155		4	21	2 (+8)

^aAdditional specimens in parentheses used for reproductive, ecological, and morphological analyses, but were not sequenced.

constant-size coalescent model. BEAST analyses ran for 2.5×10^7 generations and consensus trees were generated in TreeAnnotator 2.6.6, removing the first 20% as burnin based on Tracer plots. The resulting trees served as the input for GMYC implemented in the R package SPLITS (Ezard *et al.* 2009). The single-threshold model was used to estimate the number of candidate species under a 95% confidence interval (CI), and the ML score for delimited species compared to a null model of one species by likelihood ratio test (LRT). Results using guide trees generated under a strict-clock model were badly oversplit, estimating >30 species entities; we therefore only present results for guide trees inferred under the relaxed-clock model.

A fourth method of species delimitation employed bPTP (Zhang *et al.* 2013). A COI guide tree rooted on the outgroup taxon was generated in RaxML as described, then loaded into the bPTP server (<https://species.h-its.org/ptp/>). Analyses were run for 5×10^5 MCMC generations, discarding 25% as burnin and removing the outgroup prior to species delimitation. Outputs of both ML and BI partitions were examined to assess support for the recognized species from the Western Atlantic; acceptance rate was 0.17 for the BI partition.

Finally, *H3* alleles were mapped onto the COI gene tree alongside the corresponding COI haplotype for all individuals that yielded sequence data for both loci, to test whether nuclear alleles supported species delimitation based on COI alone and to visualize possible cases of cytonuclear discordance. Three *H3* alleles were sometimes associated with a COI haplotype sampled from multiple individuals. Phase of *H3* alleles was resolved by comparing homozygotes to the heterozygotes identified as mixed bases in chromatograms.

Ancestral character state reconstruction

To evaluate support for the hypothesis that sympatric host shifts promoted speciation in *Elysia* subclade 2, we inferred ancestral host algae for COI lineages using the BI method in BayesTraits (Pagel *et al.* 2004). Host use was modelled as a discrete variable with four possible states: (i) *Penicillus* (P), (ii) *Udotea* (U), (iii) *Halimeda* (H), or (iv) *Rhipilia* (R), based on algal associations of sampled specimens. Continuous-time Markov models were fit to host-use data to determine the posterior distribution of states at each node and transition rates, incorporating phylogenetic uncertainty by drawing a tree from the post-burnin sample of the BI analysis for each generation of the chain. Priors for ancestral state reconstructions (ASRs) and model-fitting tests were drawn from an exponential distribution generated with a hyperprior approach, with the mean of the exponential drawn from a uniform distribution (0–10). A reverse-jump (RJ) MCMC approach reduced the sampled set of model parameters by setting some transition rates to zero (Pagel and Meade 2006). MCMC chains were run for 5×10^7 iterations, discarding the first 20% as burnin, and sampling rate coefficients, host probabilities, and their associated trees every 10^5 generations.

Statistical support for an ancestral host was analysed following Kubo *et al.* (2019). The Pp distribution of ancestral states was estimated simultaneously for key nodes using the ‘addNode’ command, repeated three times, and post burnin samples were combined. A node was considered confidently reconstructed if the median Pp for one state was $\geq 50\%$, twice the null probability of equal support (25%) given four possible hosts. After ASR, branches connecting two nodes in the same state were colour-coded as retaining that state; branches connecting nodes

that differed in their inferred state were coded pink ('transitional'), to denote a historical host shift. Branches connecting a node for which state could not be confidently inferred were coded grey ('uncertain'). We also compared statistical support for alternative ancestral states using log-Bayes factor (BF) tests when no host had $\geq 50\%$ Pp at a given node by ASR. Each possible host state was sequentially fixed at that node using the 'fossil' command, and the marginal likelihood of that model was estimated using the stepping-stone sampler, running 200 stones for 2×10^5 iterations. Analyses were repeated three times for each possible host at a given node. Log-BF tests were used to compare model support, taking twice the difference between the log (L) marginal likelihood (lk) for alternative models as the BF test statistic, and comparing the median $L(lk)$ score of three runs; values >2 were taken as positive evidence (versus 5–10 as strong evidence) favouring one model over another (Raftery 1996).

Effects of larval type on population genetic structure and comparative phylogeography

We also compared levels of population genetic structure within species. We predicted species with lecithotrophic larvae would be less dispersive and show more genetic subdivision than planktotrophs, in which larval exchange maintains gene flow among demes. Then, we expected coalescent-based SDMs (GMYC, bPTP) to oversplit species with high population subdivision, compared to barcode-based SDMs (ABDG, ASAP). Development mode was only considered for the four species sampled from multiple locations and thus used in assessments of genetic subdivision. Larval type was assessed by allowing captive slugs to oviposit prior to preservation, then measuring the diameter of uncleaved ova, and the width of larval shells across the aperture at hatching (Krug 2009).

All populations of *E. patina* produced large, lecithotrophic larvae that metamorphosed quickly after hatching (Krug 2009, Krug *et al.* 2016). All *E. taino* were planktotrophic. The other two species were poecilogonous, expressing population-level differences in larval type; though rare in most invertebrates, poecilogony occurs in several sacoglossans (Vendetti *et al.* 2012, Ellingson and Krug 2016). Populations of *E. papillosa* from the southern Caribbean (Panama, Curaçao) and central Bahamas were planktotrophic (Krug *et al.* 2016), but were lecithotrophic in Florida (Clark and Goetzfried 1978) and the northern Bahamas. Most populations of *E. zuleicae* produced planktotrophic larvae, but lecithotrophic clutches were obtained from a few slugs sampled on three Bahamas islands (Sweetings Cay, San Salvador, Plana Cays) (Krug 2009, Krug *et al.* 2016).

Population genetic structure was assessed in several ways. To visualize haplotype distributions for each species, unrooted statistical parsimony networks were generated in POPART v.1.7 (Leigh *et al.* 2015) using the TCS algorithm (Clement *et al.* 2000) to infer mutational connections among extant haplotypes; the input file was a reduced COI alignment trimmed such that there were no missing data for any specimen (543 bp), coding haplotypes by sample site. Levels of population subdivision were estimated separately for each species using the same COI matrix by comparing haplotype frequencies among populations using conventional haplotype frequencies (F_{ST}) in Arlequin 2.0 (Schneider *et al.* 2000), restricting the analysis to sites with multiple sampled

individuals. AMOVA (Excoffier *et al.* 1992) was also used to assess the proportion of genetic covariance attributable to differences among sites; Φ_{ST} analyses of TrN-corrected COI distances were performed for each species, and pairwise comparisons among populations were performed when the overall test was significant.

Haplotype and nucleotide diversities were also estimated for each deme with ≥ 4 samples within each species. Tajima's D (Tajima 1989) and Fu's F_s (Fu 1997) were calculated in Arlequin 2.0 as tests of neutrality and population equilibrium for COI; significant values were interpreted as evidence of a departure from neutral expectations potentially due to a selective sweep or rapid demographic change (bottleneck or expansion; Ford 2002).

RESULTS

Evolutionary relationships

BI and ML analyses of COI haplotypes supported two major clades, each comprising multiple species with some taxa exhibiting substantial phylogeographic structure (Fig. 3). The first clade (Pp = 1.0, BS = 87%) was recovered as sister to *E. christinae*, but without significant support. Within this clade, haplotypes formed three divergent groups congruent with the described species *E. patina*, *E. papillosa*, and *E. taino*; all haplotypes of these species were $>17.7\%$ divergent (TrN-corrected COI distances) from *E. christinae*. A fully supported subclade comprised haplotypes from all sampled *E. patina* specimens, in which sequences from the Florida Keys formed a fully supported subclade up to 5.8% divergent from Bahamas haplotypes (Fig. 3). Sister to *E. patina* was a subclade (Pp = 1.0, BS = 70%) comprising the externally cryptic species *E. papillosa* and *E. taino*, supported as sister taxa. Minimum COI distance from *E. patina* to either *E. papillosa* or *E. taino* was $>14\%$.

All specimens previously identified as *E. papillosa* based on internal morphology fell within a nested subclade of haplotypes (Pp = 1.0, BS = 75%) sampled widely from across the Caribbean, including Bermuda (the type locality), Florida, the Bahamas, USVI, Jamaica, Antigua, Curaçao, and Panama (Fig. 3). Moderate phylogeographic structure was evident. Subclades of haplotypes receiving significant support were primarily or entirely sampled from one or two sites: Sweetings Cay, Little San Salvador + Bimini, Bermuda, Florida, Curaçao + Jamaica, and Panama; most haplotypes from some sites (Sweetings Cay, Bermuda) formed larger but unsupported subclades (Fig. 3). Maximum intraspecific COI distance within *E. papillosa* was 5.3%, whereas minimum COI distance to any haplotype of *E. taino* was 9.4%.

All specimens previously identified as *E. taino* by radular morphology fell within a second nested subclade of haplotypes (Pp = 1.0, BS = 83%), sampled from the southern Bahamas and central Caribbean sites (Jamaica, USVI, Dominica). Intraspecific COI distances within *E. taino* were $\leq 3.0\%$. There was some overlap in the ranges of *E. taino* and the pseudocryptic sister species *E. papillosa*. In the Bahamas, eight of nine sampled islands were inhabited by both species, but one was typically more common: *E. papillosa* comprised $>85\%$ of samples from three northern sites (Sweetings Cay, Bimini, Little San Salvador), whereas *E. taino* comprised $>70\%$ of samples from four central or southern islands (Stirrup Cay, San Salvador, Plana Cays, Compass Cay) and all samples from New Providence (Table 2; Fig. 3). Samples from

Jamaica were >85% *E. taino*, as were all samples from Dominica, yet the species was not detected in Panama or Curaçao. In contrast to *E. papillosa*, there was little evident phylogeographic structure on the COI gene tree, with only haplotypes sampled from Dominica (Pp = 1.0, BS = 92%) forming a subclade; those from all other sites formed an unresolved polytomy.

All remaining ingroup haplotypes formed a second major clade (Pp = 1.0, BS = 97%), which in turn was divided into two supported subclades. The first of those comprised four specimens from San Salvador, Bahamas, including the type specimen of *E. buonoi* (Fig. 3). Among these nominal conspecifics, maximum COI divergence was 1.4%, whereas the COI distance to any other haplotype in the clade was $\geq 10\%$. The remaining sequences formed a subclade (Pp = 1.0, BS = 94%) in which a single haplotype from the USVI was markedly divergent from the remaining specimens, which were all identified as *E. zuleicae* at the time of collection (Fig. 3). The lone divergent specimen was >11% distant from all *E. zuleicae* haplotypes, whereas maximum COI divergence within *E. zuleicae* was 3.7%. Within *E. zuleicae* there was minimal phylogeographic structure; haplotypes from the Florida Keys formed a divergent and supported (93% BS) subclade, as did some haplotypes from two Bahamas sites (Plana Cays, Bimini), but most haplotypes formed an unstructured polytomy.

Species delimitation

Excluding the outgroup, ABGD analysis recovered seven species above a prior maximal distance of 0.008, returning a barcode gap distance of 7.6% for COI (Fig. 3). Splitting to eight species required a barcode gap of only 3.2%, and split *E. patina* from the Florida Keys and Caribbean into distinct entities. The seven taxa corresponded to the six named Western Atlantic species recovered as supported subclades in phylogenetic analyses, plus the divergent USVI specimen (treated as a candidate species). The same seven species groupings were supported in the highest-scoring partition of the ASAP analysis, which returned a threshold COI distance of 6.5% (Fig. 3); the second highest-scoring partition delimited six species groups but with a threshold COI distance of 9.3%. Both ABGD and ASAP supported a large barcoding gap between 3% and 8% for this clade of *Elysia*.

In contrast to the preceding two analyses based on barcode gaps, the two multispecies coalescent SDM analyses oversplit sampled individuals based on phylogeographic structure. The effect was more pronounced for lecithotrophic species with less dispersive offspring (*E. patina*, *E. papillosa*), but was also highly influenced by the Gulf Stream, which forms a physical barrier isolating the Florida Peninsula against gene flow from surrounding Caribbean populations. GMYC analysis recovered eight species groupings when the input COI tree was generated using the relaxed clock and Yule model as priors, and the single-threshold model; this approach split *E. papillosa* into three entities but lumped the divergent individual from USVI (*E_cf_zul_100*) with *E. zuleicae* (Fig. 3). The result of this analysis was highly significant by LRT ($P = 1 \times 10^{-14}$). Within *E. papillosa*, one major cluster included individuals from sites adjacent to or north of the Gulf Stream current (Florida, Bermuda, northern Bahamas islands); another included a few Florida samples, while a third large cluster

comprised specimens from Panama, Bahamas, and Lesser Antilles sites.

GMYC analysis was more oversplit when the COI gene tree was instead generated using the constant-size coalescent model as prior, supporting 13 species entities (Fig. 3). This analysis separated the divergent USVI sample from *E. zuleicae* and *E. buonoi*, but further subdivided three other species according to phylogeographic structure. However, the two lecithotrophic species (*E. patina*: two species entities; *E. papillosa*: four entities) were no more oversplit than the planktotrophic species *E. zuleicae* (three entities). Within *E. zuleicae*, Florida samples were delimited from the larger Caribbean subpopulation, as were two specimens from Bimini, Bahamas. The split within *E. patina* was between Florida and the remaining Caribbean sites.

Under bPTP, the estimated number of species was eight under the ML partition, and between eight and 18 (mean = 8.9) in the BI partition. Five named species were supported (*E. buonoi*, *E. christinae*, *E. papillosa*, *E. taino*, *E. zuleicae*) along with the candidate species *E_cf_zul_100_USVI*, while *E. patina* was split into geographically separated Florida Keys and Caribbean lineages (Fig. 3).

Ancestral host reconstruction

Surveys at 18 sites (Fig. 2; Table 1) supported prior literature reports regarding present-day algal hosts used by sampled species. The only species recovered from *Rhipilia* was *E. christinae*. All sampled specimens of *E. patina* were recovered from *Halimeda opuntia*, while outgroup *Elysia* spp. fed on other *Halimeda* spp. Both *E. papillosa* and *E. taino* were sampled on *Penicillus* spp. (Fig. 3). Based on field associations and laboratory observations, three species used *Udotea* spp. as their primary host: *E. zuleicae*, the externally cryptic *E. buonoi*, and the candidate species from the USVI (specimen *E_cf_zul_100_USVI*).

Bayesian ASR highly supported *Penicillus* (Pp = 0.8) over three alternatives as the ancestral host for the clade comprising *E. patina*, *E. papillosa*, and *E. taino* (Fig. 3). A single ancestral host shift onto *Halimeda* was supported on the branch leading to *E. patina*. ASR also recovered almost complete Pp support for *Udotea* as the ancestral host of the subclade comprising *E. zuleicae*, *E. buonoi*, and candidate species *E_cf_zul_100_USVI* (Fig. 3). ASR returned equivocal support at deeper nodes, with Pp roughly evenly divided among two or three possible host states. Notably, none of the recent splits supported by phylogenetic analyses and SDMs were congruent with supported historical host shifts (or niche specialization from a generalist ancestor).

Cytonuclear concordance

Our range-wide phylogeographic survey yielded 234 COI haplotypes from 353 specimens (Table 2; Supporting Information Table S1). To assess the utility of a commonly sequenced nuclear marker for testing species boundaries, we plotted phased *H3* alleles onto a COI gene tree, colour-coding clades according to the best-supported taxa by SDM outputs. Candidate species *E_cf_zul_100_USVI* was omitted from this analysis due to the lack of *H3* data for this specimen. The distribution of *H3* alleles was highly congruent with species delimitation: each *H3* allele was sampled from only one of the six species on the COI tree that received

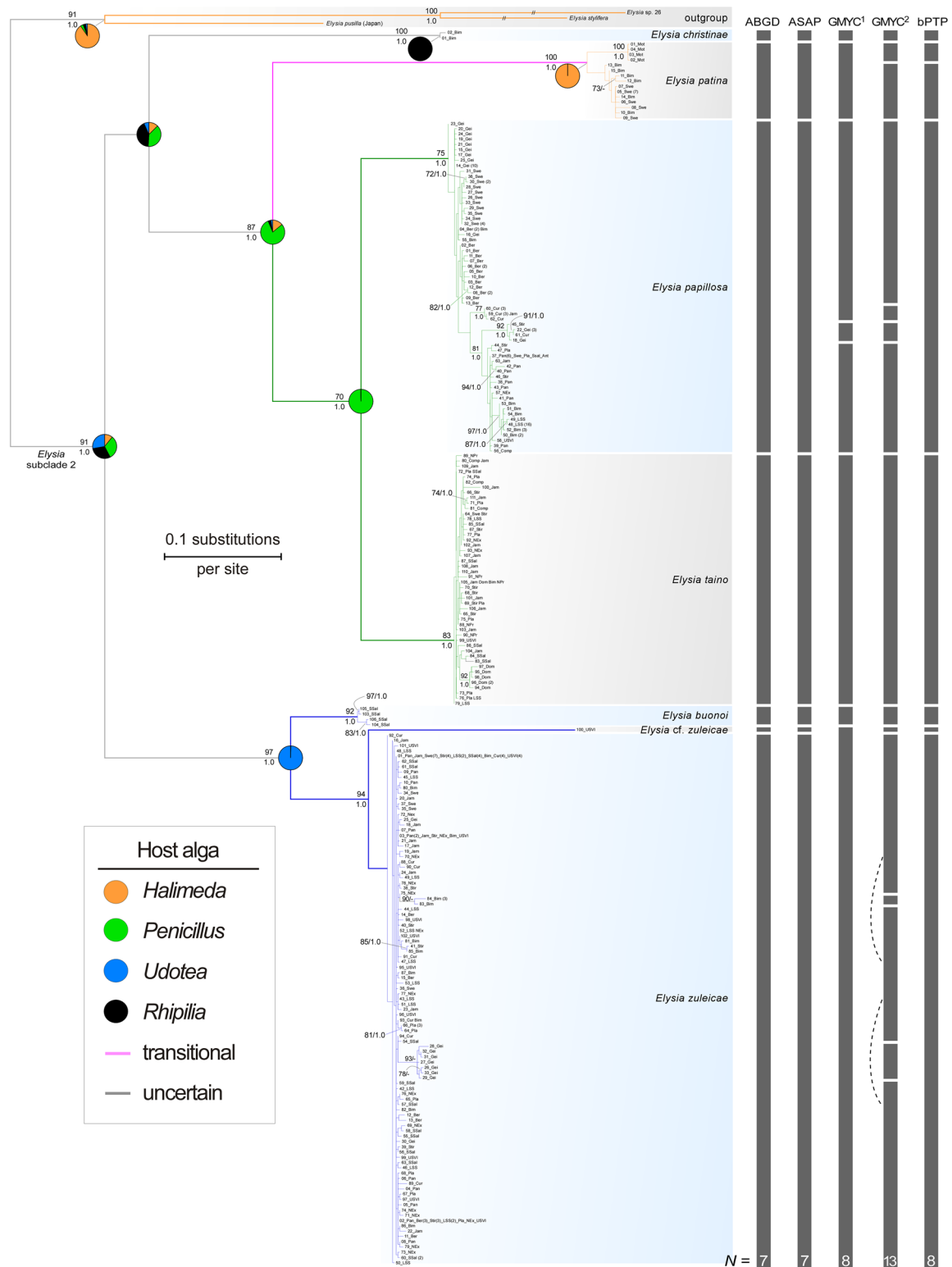


Figure 3. Evolutionary relationships and host use within the *Elysia* subclade 2 species complex. Topology and branch lengths are from the ML analysis of the *COI* gene, with Pp (above branch, before slash) or BS (below branch, after slash) indicated by numbers for nodes with significant support (dash = not significant). Colours indicate algal host, with the corresponding Pp distributions for ancestral host use given on key nodes. Branches are colour-coded as retaining their ancestral state unless connecting nodes (or a node + terminal) with different states are confidently reconstructed (>50% support for one state), in which case that branch was coloured pink for 'transitional' denoting a host shift. If the ancestral host could not be confidently reconstructed for a node, branches connecting to that node were coloured grey for 'uncertain'. Vertical grey boxes indicate species groupings receiving the highest support by five SDMs, with the total number of distinct entities given at the bottom of the box. Dashed curves join boxes representing a single species entity. For GMYC analyses, *COI* input trees were generated using a relaxed clock and Yule model (GMYC¹) or constant-size coalescent model (GMYC²) as priors.

support in at least three methods of species delimitation (Fig. 4). A given species had one to eight *H3* alleles, and multiple *H3* alleles could be associated with the same *COI* haplotype sampled from different individuals; however, the distribution of nuclear alleles largely reinforced species boundaries inferred from mitochondrial DNA (mtDNA). The only examples of cytonuclear discordance concerned three specimens from Jamaica that had mtDNA of *E. taino* but *H3* alleles otherwise sampled only from *E. papillosa*.

Population genetic structure and molecular diversity: effects of larval dispersal

Populations of *E. papillosa* showed very high and significant genetic subdivision by Φ_{ST} , with differences among sites accounting for 66% of the covariance in genetic distances (Table 3A). Nearly all pairwise Φ_{ST} estimates were also significant and most were >0.3 , with the exception of a few sites in the central Bahamas with low sample sizes (Table 4A). Overall genetic structure based on *COI* haplotype frequencies was also highly significant, with an overall F_{ST} of 24% (Table 5A); most pairwise F_{ST} comparisons were also significant, except as noted above in the central Bahamas.

In the closely related *E. taino*, populations showed less subdivision but the distribution of genetic distances among sites was still significant ($\Phi_{ST} = 0.11$, $P < .0005$; Table 3B); however, only pairwise Φ_{ST} comparisons involving Dominica were significant (Table 4B), and no genetic structure was detected by conventional F_{ST} analysis (Table 5B).

Populations in *E. patina* were exceptionally subdivided based on genetic distances among haplotypes ($\Phi_{ST} = 0.80$, $P < .00001$; Table 3C), and all pairwise comparisons among the three sampled populations were highly significant, notably those involving the Florida population MOT (pairwise $\Phi_{ST} > 0.8$; Table 4C). Genetic structure based on *COI* haplotype frequencies was also highly significant ($F_{ST} = 0.18$, $P < .005$; Table 5C); most pairwise F_{ST} comparisons were not significant due to the lack of shared haplotypes between Florida and Bahamas sites, however.

Moderately high genetic subdivision was also detected in *E. zuleicae* based on genetic distances among haplotypes ($\Phi_{ST} = 0.22$, $P < .00001$; Table 3D) and also on *COI* haplotype frequencies ($F_{ST} = 0.04$, $P < .00001$; Table 5D). More than half of pairwise comparisons were significant by either or both Φ_{ST} or F_{ST} (Table 4D).

Phylogeographic structure within species

Haplotype networks were generated to visualize the distribution of haplotypic variation among locations. In *E. papillosa*, common private haplotypes were sampled at a high frequency in the Florida Keys (GEI) and Little San Salvador, Bahamas (LSS) (Fig. 5A). A common haplotype was sampled at a high frequency in Panama and once at four other sites, and formed a starburst pattern typical of rapid expansions with haplotypes sampled from Panama and other sites. Private subnetworks without a high-frequency haplotype were evident at Sweetings Cay, Bermuda, and Curacao, all of which showed significantly negative scores by Tajima's *D* and/or Fu's F_s tests, indicating deviations from neutral expectations (Supporting Information Table S2A).

The network for *E. taino* showed no sites with a common haplotype, either private to any site or shared among sites (Fig. 5B). One haplotype was sampled once at each of four sites. There was no geographical grouping of haplotypes by site with the exception

of a subnetwork of haplotypes from Dominica. Every haplotype was unique except for two specimens from Dominica that shared a haplotype, and four of seven sites were significantly negative by Fu's F_s , consistent with recent expansion (Supporting Information Table S2B).

In *E. patina*, haplotypes from the Florida Keys (MOT) formed a subnetwork separated from Bahamas haplotypes by at least 25 mutational steps (Fig. 5C). A common, private haplotype (#05)

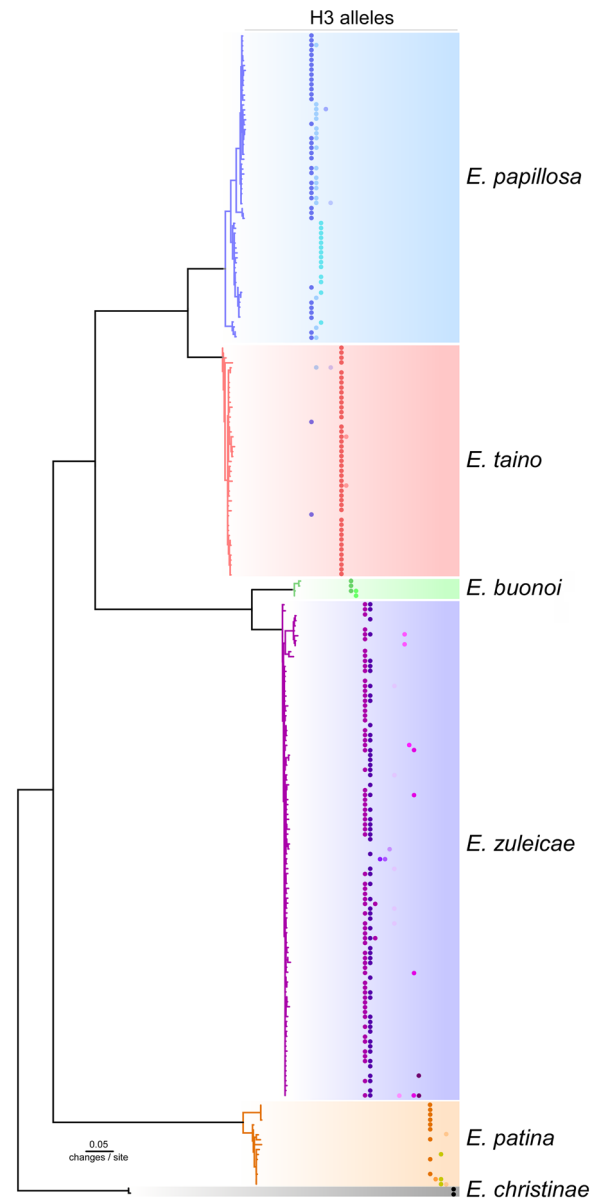


Figure 4. Cytonuclear concordance and discordance in the *Elaysia* subclade 2 species complex. Tree topology is from the ML analysis of *COI* haplotypes, pruning off outgroups and the unidentified species sampled once (*E_cf_zul_100_USVI*) for which *H3* alleles were not obtained. The six remaining taxa supported by most or all SDMs are colour-coded and in boxes. Coloured circles indicate the phased *H3* allele(s) sampled together with a given *COI* haplotype (since some individuals shared *COI* haplotypes, more than two *H3* alleles could be associated with a single *COI* sequence). Three specimens of *E. taino* had *H3* alleles (blue shaded circles) otherwise sampled only in *E. papillosa*, with which they co-occurred in Jamaica.

Table 3. Population genetic structure based on AMOVA for the *COI* locus with Φ_{ST} values computed from TrN-corrected distances, in (A) *Elysia papillosa*, (B) *E. taino*, (C) *E. patina*, and (D) *E. zuleicae*.

Source of variation	d.f	Sum of squares	Variance components	% of variation	P
A. <i>Elysia papillosa</i>					
Among populations	8	493.932	4.997	66.44	<.00001
Within populations	101	254.928	2.524	33.56	
Total	109	748.860	7.521		
B. <i>E. taino</i>					
Among populations	6	28.800	0.308	11.02	.0002
Within populations	47	117.067	2.491	88.98	
Total	53	145.868	2.799		
C. <i>E. patina</i>					
Among populations	2	98.245	7.41566	80.44	<.00001
Within populations	18	32.467	1.80374	19.56	
Total	20	130.713	9.2194		
D. <i>E. zuleicae</i>					
Among populations	12	93.873	0.509	21.56	<.00001
Within populations	140	259.337	1.852	78.44	
Total	152	353.210	2.362		

was sampled at a high frequency in Sweetings Cay, Bahamas, and formed the centre of a starburst pattern connected to haplotypes from Sweetings Cay and Bimini that were each one to seven mutational steps away. Neither the Florida Keys nor Bimini sites had a common haplotype. Sweetings Cay and the Florida Keys were significantly negative by Tajima's *D* and Fu's *F_s* tests, respectively (Supporting Information Table S2C).

The network for *E. zuleicae* was markedly different from the other species, with three common haplotypes each sampled from multiple sites, and sampled multiple times at one or more sites (Fig. 5D). The most common haplotype (#01) was sampled 30 times from 10 of 13 sites, including Curacao, the USVI, and several Bahamas Islands (Sweetings Cay, Stirrup Cay, San Salvador). This haplotype occupied the central position in a large starburst pattern that was the dominant feature of the network. A secondary starburst formed a subnetwork around the next-most common haplotype (#02) sampled 12 times from six sites. Notably, haplotype #02 was sampled in several locations (Bermuda, Panama, Northern Exumas) where haplotype #01 was not sampled, and often repeatedly at those places; unique haplotypes from those sites radiated out from the common haplotype in this subnetwork, suggesting an expanding metapopulation, despite the large geographical distances that separate Bermuda and Panama. The third subnetwork had haplotype #03 at the centre, sampled twice in Panama and once in six other places, ringed by unique haplotypes from Jamaica, Panama, and Northern Exumas. Haplotypes from the Florida Keys formed a separate subnetwork but had no common haplotype, and haplotypes from Bimini were also generally separated from the rest of the network by several mutational steps and missing sequences. Seven sites had significantly negative Tajima's *D* and/or Fu's *F_s* tests, driven by an excess of haplotypes relative to the number of variable sites; departures from neutral expectations were detected for four Bahamas islands and three

sites in the southern Caribbean (Jamaica, USVI, and Panama; Supporting Information Table S2D).

DISCUSSION

Effects of larval type and phylogeographic structure on species delimitation

Broad geographical sampling and SDM analyses support six species previously described from internal and external morphology, ecology, and development, including the disputed *E. taino* (Krug et al. 2016). We also recovered a new candidate species from the USVI, sampled once. Four species were supported by at least four out of five SDM analyses, while the others were supported by three SDM analyses. All taxa were consistently supported by barcode-based methods (ABGD, ASAP), whereas SDMs using the multispecies coalescent (GMYC, bPTP) tended to be over-split though results varied by analysis.

We predicted a gradient of population genetic structure in the four species sampled broadly across their ranges, based on their predominant larval type (Supporting Information Fig. S1). As *E. patina* expresses only lecithotrophy with short-term swimming larvae, we expected populations to be highly subdivided based on Φ_{ST} or F_{ST} values. The next-most well-supported subdivision was expected in *E. papillosa*, with lecithotrophy reported from Florida (Clark and Goetzfried 1978, Clark and Jensen 1981). Less structure was predicted for *E. zuleicae*, in which dispersive planktotrophic development predominated; lecithotrophy was only expressed by some slugs at three Bahamas sites. We predicted little structure in *E. taino*, in which all sampled demes produced planktotrophic offspring expected to drive high connectivity. Estimated subdivision based on Φ_{ST} fitted the predicted sequence (*E. patina* > *E. papillosa* > *E. zuleicae* > *E. taino*), whereas the corresponding F_{ST} values were higher for *E. papillosa* than for *E. patina*. Our

Table 4. Pairwise genetic differences among populations based on F_{ST} values calculated from conventional haplotype frequencies (above diagonal) or Φ_{ST} values based on Tamura–Nei corrected genetic distances (below diagonal), for (A) *Elysia papillosa*, (B) *E. taino*, (C) *E. patina*, and (D) *E. zuleicae*; values in bold type are significant.

A. E. papillosa													
	Swe	Stir	LSS	Central Bahamas	Bim	Gei	Ber	Cur	Pan				
Swe		0.03806	0.47751	0.05198	0.07278	0.12672	0.04167	0.12823	0.11197				
Stir	0.68459		0.72513	0.05830	0.05672	0.13006	0.01618	0.13669	0.15152				
LSS	0.89290	0.83818		0.66704	0.55482	0.50569	0.46093	0.64031	0.59423				
Central Bah	0.74776	0.08732	0.90774		0.09370	0.15586	0.05616	0.16360	−0.03655				
Bim	0.67574	0.29880	0.46690	0.30326		0.14488	0.04345	0.14900	0.16021				
Gei	0.27205	0.42867	0.74721	0.52490	0.51314		0.11051	0.19991	0.20659				
Ber	0.38665	0.74212	0.92018	0.80135	0.71603	0.29137		0.11034	0.12231				
Cur	0.63099	0.51742	0.88568	0.67750	0.62057	0.46431	0.70754		0.22132				
Pan	0.79419	0.31740	0.88338	0.01131	0.42802	0.60279	0.83650	0.76668					
B. E. taino													
	Stir	LSS	Ssal	Pla	central Bah	Jam							
Stir													
LSS	0.00888												
Ssal	0.09545	−0.11899											
Pla	−0.01868	−0.10556	0.0014										
central Bah	−0.01178	−0.08152	−0.01072	−0.06426									
Jam	0.00045	−0.08735	−0.02792	−0.03726	−0.02126								
Dom	0.49124	0.47216	0.36917	0.43749	0.4003	0.36198							
C. E. patina													
	Swe	Bim	Mot										
Swe		0.21679	0.24047										
Bim	0.23154		0.00000										
Mot	0.93132	0.83082	.										
D. E. zuleicae													
	Swe	Stir	LSS	Ssal	Pla	Nex	Bim	Gei	Ber	Jam	USVI	Cur	Pan
Swe		0.05802	0.11708	0.05931	0.25434	0.18318	0.16458	0.19838	0.25434	0.14123	0.03841	0.01828	0.15088
Stir	0.03274		0.00570	0.01314	0.09434	0.04437	0.05675	0.0708	0.0317	0.03215	−0.02493	0.00185	0.01029
LSS	−0.00132	−0.01355		0.00938	0.04302	−0.00523	0.01744	0.00777	0.01403	−0.00314	−0.00002	0.01815	−0.00511
Ssal	0.01719	−0.00712	0.01550		0.0852	0.03353	0.03291	0.03515	0.0852	0.01023	−0.01123	−0.01062	0.01933
Pla	0.21699	0.13116	0.12970	0.16185		0.04146	0.07055	0.05263	0.06323	0.05125	0.08218	0.10817	0.04995
Nex	0.03773	−0.01696	0.00341	0.02304	0.09923		0.01369	0.00000	0.0241	−0.00654	0.02752	0.05312	−0.01077
Bim	0.15425	0.10204	0.12474	0.15115	0.12495	0.08837		0.01998	0.07055	0.00561	0.02896	0.03957	0.00766
Gei	0.61203	0.58763	0.58316	0.60122	0.55102	0.55126	0.41260		0.05263	0.00000	0.0401	0.05598	0.00929
Ber	0.15895	0.02671	0.04997	0.09435	0.11259	0.02399	0.07283	0.52175		0.05125	0.06406	0.10817	0.02774
Jam	0.07732	0.02629	0.03208	0.05529	0.12085	0.00079	0.09465	0.51677	0.08631		0.00422	0.02222	−0.0161
USVI	0.03226	−0.01976	−0.01604	0.02092	0.15370	−0.01171	0.11711	0.59210	0.05613	0.02600		−0.02149	−0.00102
Cur	0.03121	0.00443	−0.00932	0.01598	0.12001	0.00083	0.10622	0.54529	0.06682	0.03936	−0.00589		0.03162
Pan	0.11460	0.01847	0.04784	0.06616	0.12660	0.01431	0.10853	0.55419	0.01891	0.03705	0.04721	0.06653	

findings thus support the prediction that larval type influences realized gene flow. The Φ_{ST} values reported here for *E. papillosa* and *E. patina* are exceptionally high for any benthic marine taxa with a planktonic stage and indicate prolonged isolation of populations.

The hypothesis that underlying population-genetic structure would bias results from SDMs was also broadly supported. Coalescent-based models oversplit lecithotrophic taxa more than predominantly (*E. zuleicae*) or entirely (*E. taino*) planktotrophic taxa, consistent with the bias of these SDMs to delimit populations rather than species (Sukumaran and Knowles 2017, Leaché

et al. 2019, Sukumaran et al. 2021). Our findings highlight the importance of sampling the range of a species with limited dispersal abilities to ensure divergent lineages are included in analyses, given the sensitivity of coalescent methods to population structure. This may be of particular concern in marine invertebrates, as development mode often varies among closely related species and profoundly alters dispersal potential. Tests of population genetic subdivision should be performed in studies that apply SDMs to marine taxa, given the potential bias introduced by larval type, to assess possible model bias during delimitation (Berriman et al. 2018, Moreno et al. 2023).

Table 5. Population genetic structure based on F_{ST} values calculated from conventional haplotype frequencies at the COI locus, in (A) *Elysia papillosa*, (B) *E. taino*, (C) *E. patina*, and (D) *E. zuleicae*; n.s., not significant.

Source of variation	d.f	Sum of squares	Variance components	% of variation	P
A. <i>Elysia papillosa</i>					
Among populations	8	14.316	0.119	23.99	<.00001
Within populations	101	38.129	0.378	76.01	
Total	109	52.445	0.497		
B. <i>E. taino</i>					
Among populations	6	3.09	0.0027	0.54	n.s.
Within populations	47	23.262	0.4949	99.46	
Total	53	26.352	0.4976		
C. <i>E. patina</i>					
Among populations	2	1.909	0.0879	18.24	.0028
Within populations	18	7.091	0.3939	81.76	
Total	20	9.000	0.4818		
D. <i>E. zuleicae</i>					
Among populations	12	8.218	0.019	3.95	<.00001
Within populations	140	64.678	0.462	96.05	
Total	152	72.895	0.481		

Our results also support the hypothesis that the Gulf Stream, the fastest western boundary current, generates genetic structure that biased coalescent SDMs to a similar degree as larval type. Most larvae entering the Gulf Stream are probably advected into the North Atlantic, isolating Florida from the Caribbean; a phylogeographic split has been detected along this current in other sea slugs (Berri-man et al. 2018, Ellingson and Krug 2016, Moreno et al. 2023). Two methods (bPTP, GMYC²) split Florida from Caribbean demes of *E. patina*. Both GMYC analyses split *E. papillosa* into one major cluster of individuals sampled along the Gulf Stream (Florida, Bermuda, northern Bahamas islands), and two or three other clusters comprising specimens from Panama, Bahamas, and Lesser Antilles sites. Pairwise genetic differentiation also supported a high degree of isolation for Florida versus the remaining Caribbean demes, where development was planktotrophic (Krug et al. 2016). In *E. zuleicae*, the GMYC² analysis also split both Florida and Bimini (located in the eastern margin of the Gulf Stream) from Caribbean demes. Thus, dispersal barriers can also generate population structure that interferes with SDM performance, and this extrinsic bias may be more consistent across co-distributed species than trait-dependent effects such as larval type (DeBiasse et al. 2016).

We also found evidence that abiotic and biotic factors can synergize to produce either highly structured or unstructured populations. Our evidence suggests larvae rarely cross the Gulf Stream; in both *E. patina* and *E. papillosa*, short-lived lecithotrophic larvae and the Gulf Stream probably reinforced the genetic isolation of Floridian subpopulations. In contrast, *E. taino* does not occur in Florida or Bermuda so its connectivity matrix does not reflect interruption by this major current system, instead exhibiting high gene flow among demes mediated by planktotrophic larvae. Moreover, all haplotypes were unique in *E. taino*, which can cause conventional F_{ST} analysis to underestimate subdivision (Jost et al. 2018, Alcalá and Rosenberg 2019). In this species, multiple factors thus synergize to produce cohesion among demes.

Algal host shifts and ecological divergence

Ecotypes may undergo divergence with gene flow even in marine systems lacking obvious dispersal barriers, suggesting disruptive selection may be an underappreciated driver of speciation (Hurt et al. 2013, Johannesson et al. 2017, Momigliano et al. 2017, González et al. 2018, Teske et al. 2019, Leder et al. 2021, Fritts-Penniman et al. 2020). However, formal methods to identify the effects of trait change on cladogenesis are generally lacking (Goldberg and Igić 2012). Although host preference can inform species delimitation (Galen et al. 2018), to our knowledge, there is no established method for testing event-dependent diversification, that is, whether host shifts (change in a multistate character) trigger speciation.

The diversity of hosts used in *Elysia* subclade 2 exceeds that in most of the recognized subclades within *Elysia* (Lugo et al. 2025), suggesting an unusually high degree of host switching in this radiation. ASR modelling supported two possible host shifts associated with speciation events. The shift onto *Rhipilia* in the ancestor of *E. christinae* may have been a driver of speciation, especially as no other sacoglossan is known to specialize on this alga. ASR also supported sequential shifts first onto *Penicillus*, then back to *Halimeda* (the host of the outgroup) in the ancestor of *E. patina*. Most *Halimeda*-feeding elysiids are not restricted to a single host species, but *E. patina* was sampled exclusively from *H. opuntia*, making it one of the most stenophagous marine herbivores.

In contrast, the shift onto *Penicillus* occurred in the ancestor of *E. papillosa* + *E. taino* and thus preceded speciation in this lineage. Both descendant slug species feed on *Penicillus*, but may prefer different algal species (Krug et al. 2016, Middlebrooks et al. 2019, Barber et al. 2021). Similarly, the shift onto *Udotea* pre-dated the radiation of three *Elysia* species that use this alga as host. Thus, in this radiation, several host shifts preceded recent speciation events. This is consistent with phylogenetic evidence that most host shifts pre-date recent radiations in *Elysia*, in which

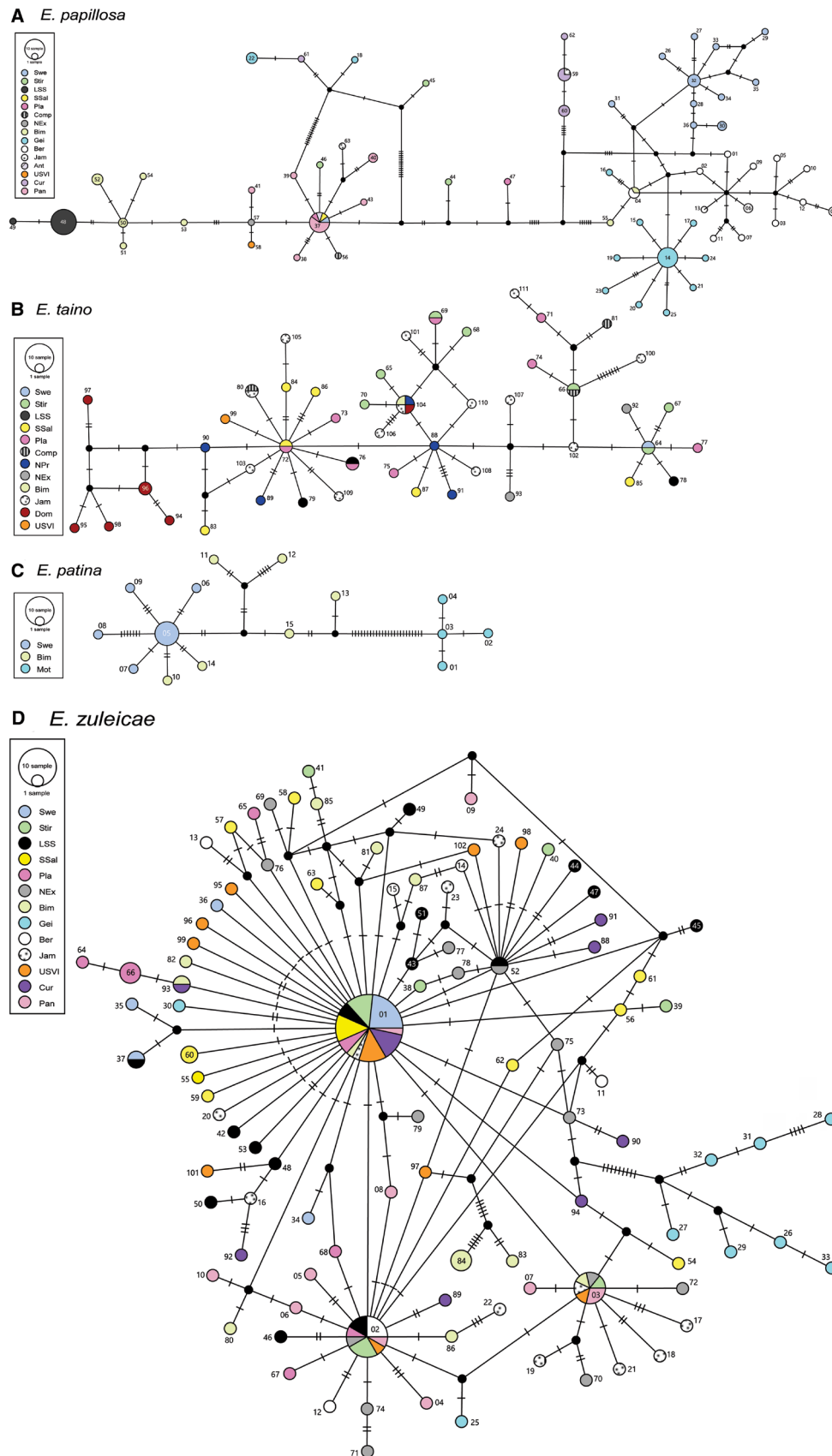


Figure 5. Statistical parsimony networks of *COI* haplotypes from four species in the *Elaysia* subclade 2 species complex. Networks were inferred under a 95% parsimony criterion. Circle area is proportional to the frequency of each haplotype, colour-coded by sampling site and numbered (Supporting Information Table S1). Black dots represent unsampled haplotypes. Slash marks across a line segment indicate the number of substitutions. A, *E. papillosa*. B, *E. taino*. C, *E. patina*. D, *E. zuleicae*.

host use is conservative (Lugo *et al.* 2025). Given the long branches leading to *E. christinae* and *E. patina*, there may be unsampled sister taxa on the same hosts, potentially in the East Atlantic (Lugo *et al.* 2025). Overall, host shifting appears to pre-date most recent splits in *Elysia*, but the present study highlights some additional cases of potential ecological speciation (Rodríguez and Krug 2022). It remains unclear whether ecological specialization and physical barriers to dispersal reinforce to limit gene flow in *Elysia*, given the timing of host shifts in this radiation, but other recent studies provided some evidence that these factors synergize to promote divergence (Moreno *et al.* 2023).

Cytoneuclear concordance and transient allopatry

Divergence between *E. papillosa* and *E. taino* was probably driven by Pleistocene sea level fluctuations and transient allopatric isolation in separate parts of the Caribbean basin. These species remain ecologically and morphologically very similar, but are distinct in their radular tooth morphology. The ranges are partially overlapping, but *E. papillosa* is the predominant species in most northern sites, as well as along the northern coast of South America; *E. taino* predominates in the central Caribbean and eastern Lesser Antilles islands. Their distributions suggest they entered into secondary contact following sea level rise and population expansion, but each species may be competitively excluding the other from most of its historical range.

Despite overlapping in most Bahamas sites and Jamaica, we found no evidence for F1 hybrids, which would have one *H3* allele from each parent species. Three specimens exhibited cytonuclear discordance, with mtDNA that grouped phylogenetically within *E. taino*, but one was a heterozygote for two common *E. papillosa* alleles, and the other two were homozygotes for a common *E. papillosa* allele. These specimens probably reflect backcrossed offspring of hybrids that mated with *E. papillosa*. Indeed, the Jamaica population of *E. papillosa* may have undergone ‘mitochondrial capture’ with some specimens now carrying mtDNA of *E. taino* due to asymmetric introgression; this phenomenon was recently documented in another pair of Caribbean elysiids (Rodríguez and Krug 2022). While our sampling in Jamaica indicated that *E. papillosa* was locally rare ($N=2$) compared to *E. taino* ($N=13$), those findings were based on *COI* sequences. The *H3* data suggest that *E. papillosa* may be more abundant in Jamaica but could barcode as *E. taino* due to introgressed mtDNA. Mitochondrial capture is rare among internally fertilizing marine invertebrates, but potentially more common than currently recognized given recent examples (Darling 2011, Toews and Brelsford 2012, Layton *et al.* 2020, Rodríguez and Krug 2022).

Morphological investigation and/or genome-wide analyses would help to resolve the contributions of the two parent species to the discordant individuals identified here. Data from one nuclear locus was a valuable complement to mtDNA-based barcoding, helping to delimit species and identifying cases of potential hybridization or introgression. However, genome-wide data would provide more powerful tests of admixture and permit delimitation analyses that explicitly incorporate evolutionary processes (Smith and Carstens 2020). Such approaches are starting

to be more widely applied to marine gastropods, including the use of single-nucleotide polymorphisms (Simmonds *et al.* 2020) and target-capture enrichment approaches (Teasdale *et al.* 2016) that may facilitate studies of host shifting versus alternative models of divergence with gene flow.

Sexual selection and divergence

Hypodermic insemination occurs in a number of sacoglossan genera, but is relatively uncommon in *Elysia*; subclade 2 and our outgroup, subclade 1, are among the few lineages in which penial stylets occur in *Elysia* (Gascoigne 1974, Lugo *et al.* 2025). It was previously noted that *E. zuleicae* has a stylet, but *E. buanoi* does not (Krug *et al.* 2016). The latter species has still only been sampled from a single Bahamas island. Sexual selection on penial armature in sacoglossans has been suggested to promote rapid reproductive isolation and local endemic species, especially acting in concert with shifts to non-dispersive larvae and in host use (Moreno *et al.* 2023, 2025). Lack of material prevented a full description or morphological investigation of the new species detected here from the USVI from a single specimen. However, we hypothesize that this species, also a *Udotea*-feeder provisionally identified as *E. zuleicae*, will show a marked change in stylet morphology or independent loss of the stylet, like *E. buanoi*. If supported, this hypothesis suggests endemic species can rapidly form through loss of penial armature. Because these species are rare relative to their stylet-bearing relative, they are likely to have much smaller effective population sizes, another parameter that can bias the outcomes of coalescent-based SDMs. Notably, one coalescent method did not recover the USVI specimen as a distinct species in contrast to the other methods, suggesting some aspect of this rare lineage impeded successful delimitation. Identifying the mechanisms responsible for the production of rare and endemic species, and how they affect SDM performance, are thus important objectives for species discovery research.

Systematic implications

Our range-wide sampling data and SDM results support prior taxonomic hypotheses for six *Elysia* spp. in the Western Atlantic that were distinguished by sequence data, host ecology, radular and reproductive morphology, and larval type (Krug *et al.* 2016). The systematic status of these species has been confused and contentious, partly due to incomplete or misinterpreted descriptions of *E. papillosa* (type locality: Bermuda) and *E. patina* (type locality: Florida, USA). Ortea Rato *et al.* (2005) shuffled these names despite prior systematic continuity (Marcus and Marcus, 1967, Marcus 1980, Clark 1984), though subsequent work clarified the species identities (Gowacki *et al.* 2016, Krug *et al.* 2016, Rodríguez and Krug 2022). Ortea Rato *et al.* (2005) re-described ‘*E. papillosa*’ using material probably comprising *E. pawliki* and other species from Costa Rica, Cuba, and the Canary Islands, none of which are known from Bermuda (Rodríguez and Krug 2022). Ortea Rato *et al.* (2005) also re-described ‘*E. patina*’ using specimens that were not *E. patina* Ev. Marcus 1980, but probably *E. papillosa* based on the teeth illustrated (Ortea Rato *et al.* 2005: fig. 4D), resulting in a name swap (Gowacki *et al.* 2016, Krug *et al.* 2016). Finally, Ortea Rato *et al.* (2005) described *E. annedupontae* from

Cuba with the diagnostic features of the radula, penial anatomy, and rhinophores that Marcus and Marcus (1967) identified in *E. papillosa*. Having in the same paper assigned the name '*E. papillosa*' to a different congener, Ortea Rato *et al.* (2005) thus gave the name *E. annedupontae* to *E. papillosa sensu* Verrill (1901), without explaining how their '*E. patina*' differed from *E. annedupontae*; neither is mentioned in the remarks for the other, despite indistinguishable descriptions. Krug *et al.* (2016) identified *E. annedupontae* as a junior synonym of *E. papillosa*, and '*E. patina*' *sensu* Ortea Rato *et al.* (2005) as mis-identified *E. papillosa*.

Elysia taino was described as superficially similar to *E. papillosa* and also feeding on *Penicillus*, but distinct by molecular data and radular morphology (Krug *et al.* 2016). Measuring from scanning electron microscopy (SEM) images, the leading tooth of *E. papillosa* ($N=18$) is narrow and blade-like, $130 \pm 11 \mu\text{m}$ long (mean $\pm$ SE) by $16 \pm 2 \mu\text{m}$ wide at the halfway point of the serrated edge (width:length ratio of 0.12 ± 0.01). In contrast, the leading tooth in *E. taino* ($N=6$) is significantly longer and wider ($189 \pm 22 \mu\text{m}$ long by $30 \pm 3 \mu\text{m}$ wide at the halfway point), and proportionally thicker (width:length ratio of 0.16 ± 0.01). As shown here, the taxa rarely hybridize even where co-occurring, and we detected no other *Penicillus*-feeding species in 20 years of intensive sampling. Ortea (2018) declared *E. taino* a junior synonym of *E. annedupontae*, without molecular data and without visiting the type locality or examining type material. The description of *E. annedupontae* included a drawing of the radular tooth that measured $\sim 150 \mu\text{m}$ long by $17 \mu\text{m}$ wide (width:length ratio = 0.11; Ortea Rato *et al.* 2005: fig. 3D). Those measurements are within the range of *E. papillosa* but outside the range of *E. taino*, supporting *E. annedupontae* as a junior synonym of *E. papillosa* (Krug *et al.* 2016). Arguing that *E. annedupontae* was a senior synonym of *E. taino*, Ortea (2018) redrew the tooth of the holotype but with altered dimensions ($175 \mu\text{m}$ long by $23 \mu\text{m}$ wide), a more angled morphology, and a width:length ratio of 0.13 (Ortea 2018: plate 5D). Although the modified proportions make the tooth look more like that of *E. taino*, even as redrawn, the tooth is too thin to be from *E. taino*. Notably, the tooth of '*E. patina*' was also redrawn with very different proportions to make it resemble that of *E. papillosa*, whereas the original tooth illustrations from '*E. patina*' and *E. annedupontae* in Ortea Rato *et al.* (2005) are indistinguishable (compare Ortea Rato *et al.* 2005: fig. 4G with Ortea 2018: plate 5B). No anatomical or molecular data support the hypothesis that the holotype of *E. annedupontae* is conspecific with *E. taino*. We restore *E. taino* and again recognize *E. annedupontae* as a junior synonym of *E. papillosa* (see also Krug *et al.* 2016).

In re-litigating the status of *E. papillosa* versus *E. annedupontae*, Ortea (2018) acknowledged the similarities between *E. annedupontae* and Verrill's description of *E. papillosa*. However, Ortea (2018) asserted the radular tooth illustrated for *E. papillosa* by Marcus or the Marcuses is clearly distinct from radular SEM images of *E. papillosa* in Krug *et al.* (2016). A redrawn tooth (Ortea 2018, plate 5E) is mis-attributed to Marcus (1980), which does not illustrate a radula for *E. papillosa*, but is presumably the tooth from Marcus and Marcus (1967: fig. 23). Ortea (2018) reports the Marcuses' tooth of *E. papillosa* is 'idéntico al de *Elysia taino*'

(identical to that of *E. taino*), but the lone specimen examined by Marcus and Marcus (1967) was from Biscayne Bay, Florida, USA; as *E. papillosa* occurs in Florida but *E. taino* does not, Marcus and Marcus (1967) described *E. papillosa sensu* Verrill (1901). Ortea (2018) asserts that *E. papillosa* must have two brown bands on its rhinophores. Most *Elysia* spp. in subclade 2 have one to two dark bands across the rhinophores (Fig. 1), with the distal band varying in darkness within and among species. Rhinophore shape is also deformable and flexible, so these are not diagnostic criteria in this subclade.

There is only one species in Bermuda that conforms to the original description of *E. papillosa*, to what other authorities recognized as *E. papillosa* prior to 2005, and to what other authorities have recognized as *E. papillosa* since 2016. Confoundingly, Ortea (2018) proposed a new name (*Elysia frankenstein*) for the species Ortea Rato *et al.* (2005) called *E. patina* in 2005, and which all other cited authorities identified as *E. papillosa* Verrill 1901 (Marcus and Marcus 1967, Redfern 2013, Gowacki *et al.* 2016, Krug *et al.* 2016). No anatomical details were provided beyond a drawing of the head, a photograph of a typical *E. papillosa* specimen, and statements about the teeth and a penial stylet, all consistent with *E. papillosa*. We synonymize *E. frankenstein* with *E. papillosa* (Gowacki *et al.* 2016, Krug *et al.* 2016).

We also identified two species similar to *E. zuleicae*, the recently described *E. buonoii* and the undescribed specimen from the USVI, suggesting *E. zuleicae* shares a common ancestor with several endemic species. Ortea Rato *et al.* (2005) described *E. deborahae* as a species very similar to *E. zuleicae*, but noted supposedly diagnostic differences (absence of a tail, presence of a penial stylet, differences in shape and number of radular teeth). Krug *et al.* (2016) concluded those differences fell within the intraspecific variation for *E. zuleicae*. Similarly, *Elysia sarasuae* Ortea and Espinosa in Ortea, Moro, Caballer and Espinosa 2011 is unrecognizable from available data as other than a small *E. zuleicae*. Finally, the description of *Elysia leanneae* Caballer, Ortea and Espinosa 2013 did not provide details outside the range of intraspecific variation noted for specimens of *E. zuleicae* here and in prior studies (Krug *et al.* 2016). We thus consider *E. deborahae*, *E. sarasuae*, and *E. leanneae* as junior synonyms of *E. zuleicae*, without molecular or anatomical evidence of diagnostic traits supporting their status as distinct taxa. Careful analysis is needed to differentiate rare species from more common relatives that exhibit high intraspecific variation, but such taxa (especially if endemics) may warrant conservation measures (Moreno *et al.* 2023).

CONCLUSIONS

More recent theory and the ever-increasing richness of genomic datasets has led to a new generation of SDMs that incorporate a gradual speciation processes, minimizing known artefacts and delimiting species consistent with broader taxonomic patterns (Dellicour and Flot 2015, Luo *et al.* 2018, Leaché *et al.* 2019, Sukumaran *et al.* 2021). New process-based SDMs will allow explicit model-fitting and the incorporation of data drawn from ecology, geography, or hybrid zones between candidate taxa (Smith and Carstens 2020). Such studies bridge biodiversity

discovery programmes with eco-evolutionary theory, revealing mechanisms that promote or constrain reproductive isolation. As SDMs become increasingly sophisticated, it should be possible to test multiple processes acting at different stages during model-fitting, and to delimit the most biologically meaningful groups. Traits that inform taxonomic investigation can thus be incorporated into SDMs to inform delimitation, but also to test when different evolutionary forces act during a radiation.

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AUTHOR CONTRIBUTIONS

Patrick J. Krug (conceptualization, funding acquisition, data curation, investigation, formal analysis, resources, supervision, visualization, writing—original draft, review & editing), Danielle Y. Trathen (investigation, formal analysis), Jennifer Retana (investigation, formal analysis), Diane M. Rico (investigation, formal analysis), Julius Bronolo (investigation, formal analysis), and Ryan A. Ellingson (conceptualization, funding acquisition, investigation, formal analysis, resources, supervision).

SUPPLEMENTARY DATA

Supplementary data are available at *Biological Journal of the Linnean Society* online.

CONFLICT OF INTEREST

The authors declare no conflicts of interest.

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DATA AVAILABILITY STATEMENT

DNA sequences: NCBI accession numbers are given in [Supporting Information Table S1](#). DNA alignments, Arlequin input files, and output from *BayesTraits* ancestral host reconstructions are archived on Dryad (10.5061/dryad.0gb5mkmff).

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